Syllabus: Philosophy of Science

Instructor: Aydin Mohseni
Email: amohseni@uci.edu
Office: SST 786
Office Hours: Thursdays 1-3pm

Course Description

Science constitutes the most successful human endeavor in cumulative acquisition of knowledge of the natural world. Why? What, if anything, separates science from pseudoscience? How do theories change? How literally should we interpret our best physical theories? How can evidence underdetermine the correct theory? What does scientific explanation consist in? What are the roles of models in science? Can we precisely formulate the relationship of evidence to theory? Why is a simpler theory better? And how are we to understand the current replication crisis in the social and life sciences? Together, we'll explore these questions and others as we do weekly readings, discuss, and embark together on a path to a richer understanding of the philosophy of science.

Student Learning Outcomes

By the end of this course, you will be able to:

1. Summarize some of the central debates in philosophy of science.
2. Evaluate the arguments and evidence for the various positions in these debates

Course Texts

You do not need to purchase any texts for this course. All readings will be made available on the course site.

Assignments

Participation: you are expected to actively participate in the class discussion by bringing your own questions, listening and responding to the questions of your peers, and offering your own points of view.

Reading summaries: periodically, you will be asked to submit concise reading summaries demonstrating your understanding of the texts.

Exams: there will be one midterm exam and one final exam. The final exam will be cumulative. Final essay: you will be asked to write a 1000-1500 word essay on either

(A) a theme in one of our three texts, or (B) a topic of connection between two of our texts.

Extra Credit. There will be one opportunity to earn extra credit by writing a short (600-1,000 words) reflection essay about how you see the material we cover in class connecting to your life, studies, and work, and to the development of your intellectual character.

Grading Breakdown

You final grade for the course will be determined by:
Participation - 15%
Reading summaries - 15%
Midterm - 20%
Final Exam - 25%
Final Paper - 25%

Grading Scale

A+ : 100-98 $\geq$ A : 97-93 $\geq$ A- : 92-90 $\geq$
B+ : 89-88 $\geq$ B : 87-83 $\geq$ B- : 82-80 $\geq$
C+ : 79-78 $\geq$ C : 77-73 $\geq$ C- : 72-70 $\geq$
D+ : 69-68 $\geq$ D : 67-63 $\geq$ D- : 62-60 $\geq$ F : 59-0

Late Work Policy

Late work will not be accepted.

Academic Honesty

This course will follow the University policy on academic integrity.

Course Calendar

Below are the readings that should be completed by beginning of each class. All readings will be made available on the course website.

WEEK 1. **Falsification & Testabilty**
Karl Popper (1963) "Science as Falsification" in *Conjectures and Refutations*
Elliott Sober (1999) "Testability" *Proceedings and Addresses of the American Philosophical Association,* 73 (2):47-76

WEEK 2. **Theory Change**
Thomas Kuhn (1962) "Progress through Revolutions" in *The Structure of Scientific Revolutions*
Imre Lakatos (1978) "A methodology of scientific research programmes" in *Methodology*

of Scientific Research Programmes

WEEK 3. **Realism and Antirealism**
Laudan (1981) "A Confutation of Convergent Realism" *Philosophy of Science*, 48 (1):19-49
Kyle Stanford (2019) "A Middle Path Forward in the Scientific Realism Debate"

WEEK 4. **Confirmation Holism**
Duhem, Pierre (1954) "To Save the Phenomena" in *The Aim and Structure of Physical Theory*
Deborah Mayo (1997) "Duhem's Problem, the Bayesian Way, and Error Statistics" *Philosophy of Science*, 64:222–44

WEEK 5. **Scientific Explanation**
Carl Hempel and Paul Oppenheim (1948) "Studies in the Logic of Explanation" *Philosophy of Science*, 15 (2):135-175
James Woodward ()2007) "Causation with a Human Face" in *Causation, Physics, and the Constitution of Reality* pp. 66–105.
Bas Van Fraasen, (1980) "To Save the Phenomena," Ch. 3 in *The Scientific Image*

WEEK 6. **Scientific Models**
Ludwig Boltzman (1911) "Model" *Encyclopedia Britannica.* Transcribed version.
Ron Giere (2004) "How Models are used to Represent Reality," *Philosophy of Science*, 71:742–752.

WEEK 7. **Scientific Idealization**
Ernan McMullin (1985) "Galilean Idealization" *Studies in History and Philosophy* 247-273.
Michael Weisberg, "Three Kinds of Idealization," *The Journal of Philosophy*, 104 (12):639-659.

WEEK 8. **MIDTERM**

WEEK 10. **Measures of Confirmation**
James Joyce, "Baye" Theorem" in *The Stanford Encyclopedia of Philosophy*
Colin Howson and Peter Urbach, 11Bayesian vs Non-Bayesian Approaches to Confirmation," from Ch. 7 of *Scientific Reasoning: The Bayesian Approach.*

WEEK 11. **Scientific Virtues**
Thomas Kuhn (1977) "Objectivity, Value Judgment, and Theory Choice," Ch. 13, pp. 320-339 in *The Essential Tension*
Heather Douglas (2013)"The Value of Cognitive Values," *Philosophy of Science*, 80:796-806.

WEEK 12. **The Social Structure of Science I**

Philip Kitcher (1990) 'The Division of Cognitive Labor" *The Journal of Philosophy* lxxvii (1):5–22

Michael Strevens (2006) "The Role of the Priority Rule in Science" *Journal of Philosophy*, 2:55–79

WEEK 12. **The Social Structure of Science II**

Mayo-Wilson, Kevin J.S. Zollman, and David Danks (2011) "The Independence Thesis: When Individual and Social Epistemology Diverge" *Philosophy of Science* lxxviii, 4:653–77.

Kevin Zollman (2018) "The Credit Economy and the Economic Rationality of Science" *Journal of Philosophy,* 115 (1):5-33

WEEK 13. **The Replication Crisis I**

John Ioannidis (2005) "Why Most Published Research Findings Are False" *PLoS Med* 2(8)

False-Positive Psychology (2011) "False-Positive Psychology: Undisclosed Flexibility in Data Collection and Analysis Allows Presenting Anything as Significant" *Pyschological Science* 11:1359-1366

WEEK 14. **The Replication Crisis II**

—*Final Paper Due*—

Remco Heesen (2018) "Why the Reward Structure of Science Makes Reproducibility Problems Inevitable" *The Journal of Philosophy*, 115 (12):661-674

David Grimes, Chris Bauch, and John Ioannidis (2017) "Modeling science trustworthiness under publish or perish pressure" *Royal Society Open Science*

WEEK 15. **FINAL EXAM**

Syllabus: Introduction to Inductive Logic

Instructor: Aydin Mohseni
Email: amohseni@uci.edu
Office: SST 786
Office Hours: Tuesdays 2-4pm

Course Description

All knowledge of the natural world is attained by way of inductive inference. But some methods of inductive inference are better than others. This difference is of enormous importance, and this course aims to provide you with an understanding of what constitutes this difference.

What determines the difference between reliable and unreliable inductive methods? How, if at all, is induction justified? These question have implications for how we approach predicting and understanding everything from effective medical treatments, successful social policies, to climate change and collective and individual human behavior, and much else besides. Indeed, inductive inference is central to every human activity aimed at discovering reliable patterns for prediction and manipulation of the natural world. Together, we'll explore these questions on the path to understanding the power and pitfalls of inductive logic.

Student Learning Outcomes

By the end of this course, you will be able to:

1. Identify and correct for persistent cognitive biases in inductive reasoning.
2. Demonstrate how to update one's beliefs using Bayes' rule, and to apply this to problems in medicine, law, and human behavior.
3. Represent causal and probabilistic arguments using Bayes' nets.

Course Texts

You do not need to purchase any texts for this course. All readings will be made available on the course site.

Assignments

Weekly Problems Sets. There will be a weekly problem set. The completed homework will be due at the beginning of each Monday. This is meant for you to deepen your understanding of the readings and material from class.

Daily Questions. At the beginning, middle, and end of each of each class, you will be asked to answer a question prompt. The question at the beginning of class will be on the assigned readings. If you have done the readings, you should be able to answer the question. Questions during the class to see how well/whether you are following the lesson, and will be graded mainly for completion. The question at the end of class will be on the material covered during that class. If you have paid close, critical attention during class, you will do well.

Midterm Exam. The midterm examination will be a comprehensive review of the content from first half of the course. It will cover the foundations of probability theory, the Kolmogorov axioms, applications of Bayes' rule, and the coherence arguments: Dutch book and calibration. A review session will be held before the midterm.

Final Project. There is no final exam. Instead, there is a final project. For the final project in this course, you assemble a presentation, providing a causal-probabilistic analysis of an actual case study. The case study can be from law, medicine, human behavior, or wherever probabilistic reasoning may be fruitfully applied. In doing so, you will employ all the principals you have learned in the course on effective inductive reasoning to argue what our degrees of belief should be concerning the primary hypotheses in the case in question.

How to Succeed in this Course

Learning inductive logic is much like learning any new language. It is not enough to grasp the grammar of the language in the abstract—you must internalize it through repeated practice and application. You want to become conversant in inductive logic. To this end, the weekly assignments are meant to helping develop fluency with the concepts learnt in class. Don't skip them thinking you that have "understood the concepts" through mere exposure in class or in the readings.

Moreover, it has been thoroughly demonstrated that the most effective way of learning is teaching. You are free to collaborate with others on all of the homework, though you must write up your own work individually. I encourage students to engage one another in study groups. Practice explaining your understandings to your classmates, and indeed to anyone who listen.

Attend all of the classes, come prepared to engage, do all of the homework, and practice explaining the concepts you are learning to others. Do all of this, and you will do great in the class.

Grading Breakdown

You final grade for the course will be determined by:
Final Project - 30%
Midterm Examination - 30%
Daily Questions - 20%
Assignments - 20%

Grading Scale

A+ : 100-98 $\geq$ A : 97-93 $\geq$ A- : 92-90 $\geq$
B+ : 89-88 $\geq$ B : 87-83 $\geq$ B- : 82-80 $\geq$
C+ : 79-78 $\geq$ C : 77-73 $\geq$ C- : 72-70 $\geq$
D+ : 69-68 $\geq$ D : 67-63 $\geq$ D- : 62-60 $\geq$ F : 59-0

Late Work Policy

Late work will not be accepted.

Academic Honesty

This course will follow the University policy on academic integrity.

Course Calendar

Below is a table indicating readings and assignments that are due at the beginning of each class. All readings will be made available on the course website on Canvas. The following abbreviations are employed throughout the course calendar:

IL: *Choice and Chance: and Introduction to Inductive Logic* by Brian Skyrms.
FS: *Thinking Fast and Slow* by Daniel Kahneman
SF: *Super Forecasting* by Philip Tetlock and Dan Gardner
SN: *The Signal and the Noise* by Nate Silver
LR: *Games and Decisions* by Duncan Luce and Howard Raiffa
JP: *Causality* by Judea Pearl

	LESSON	READINGS	HW
WEEK 1	Intrdouction & Motivation		
WEEK 2	Old and new problems of induction and meta-induction	IL: Ch. 1	HW1
WEEK 3	Kolmogorov axioms: an intro to probability theory	IL: Ch. 2	HW2
WEEK 4	Bayes' theorem: learning from evidence	IL: Ch. 3, 4	HW3
WEEK 5	Bayes' theorem: total evidence and reflection	IL: Ch. 5	HW 4
WEEK 6	Bayes' theorem: applications to law and medicine	IL: Ch. 8	HW5
WEEK 7	Heuristics & biases: base rate neglect, conjunction fallacy, salience bias, and regression to the mean	FS: 10,12,17	HW6
WEEK 8	MIDTERM EXAM		
WEEK 9	Expert judgment: learning better and worse	SF: 3, SN: 4,5	HW7
WEEK 10	Coherence arguments: Dutch book, accuracy, value of knowledge, and representation theorems	IL: Ch. 9	HW8
WEEK 11	Expected utility theory: the logic of decision	LR: Ch. 2	HW9
WEEK 12	Introduction to causal modeling using Bayes' nets	JP: Ch. 1	HW10
WEEK 13	Applying Bayes' nets to case studies	JP: Ch. 6	HW11
WEEK 14	Further applications of causal-probabilistic reasoning	JP: Ch. 7	HW12
WEEK 15	FINAL PROJECT		